



Project Site

FlatLands

Generative Floormap Completion from a Single Egocentric View

Subhransu S. Bhattacharjee • Rahul Shome • Dylan Campbell

School of Computing • The Australian National University

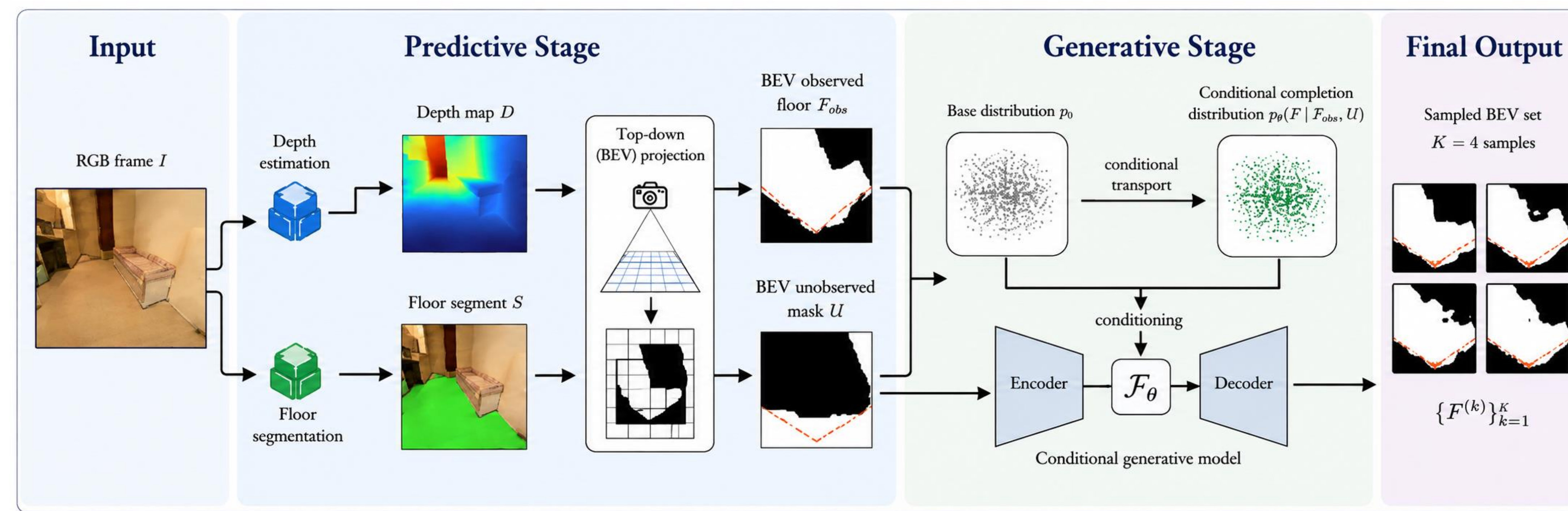


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Datasets & Benchmarks
Computer Vision for Robotics

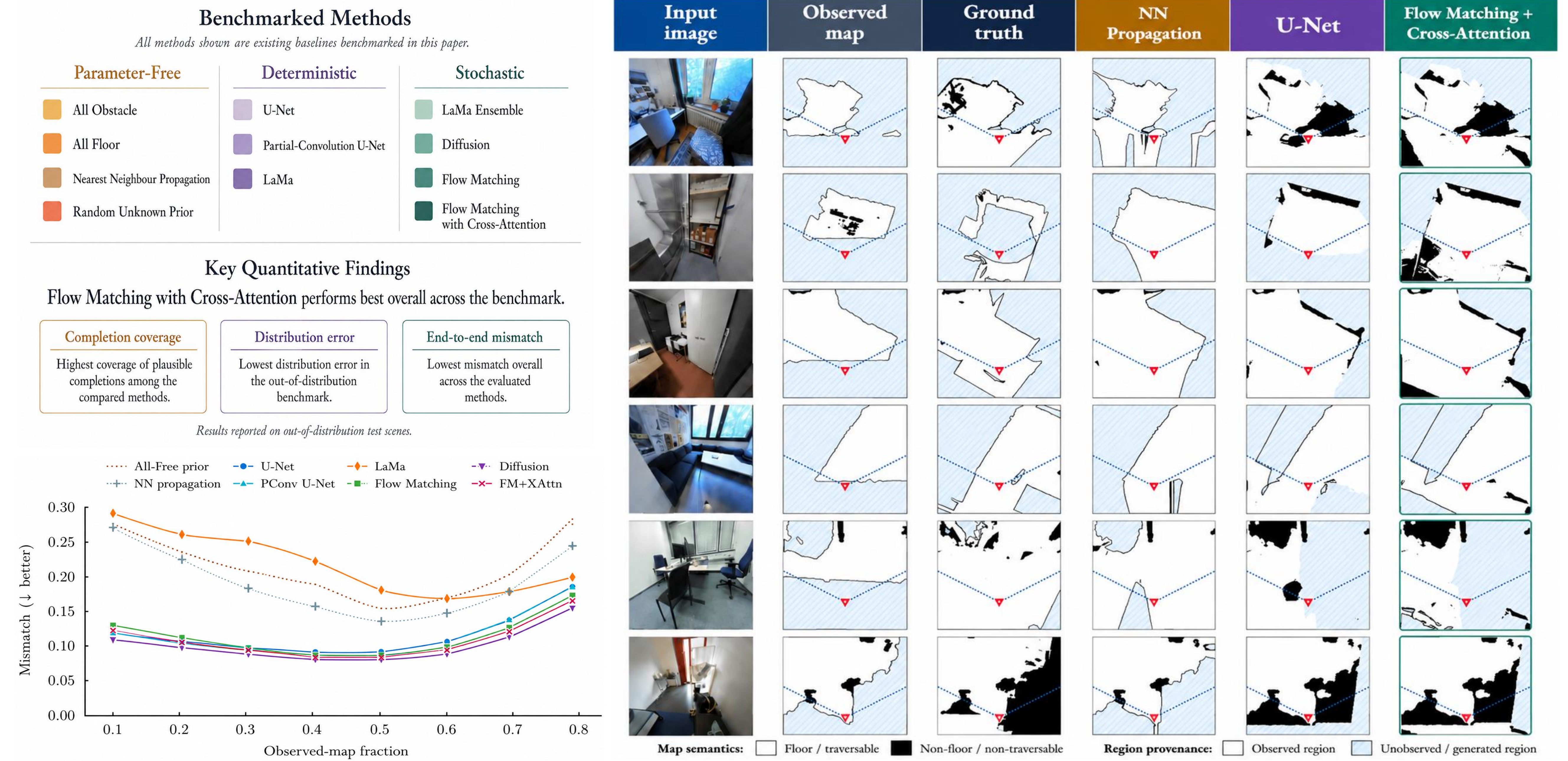
Floormap Completion

- Problem:** Predicting a complete bird's-eye-view traversability map from one egocentric RGB image to reason beyond the field of view of the visible scene.
- Challenge:** Unobserved regions admit multiple plausible layouts, motivating conditional 2D generative completion with deterministic and stochastic baselines.

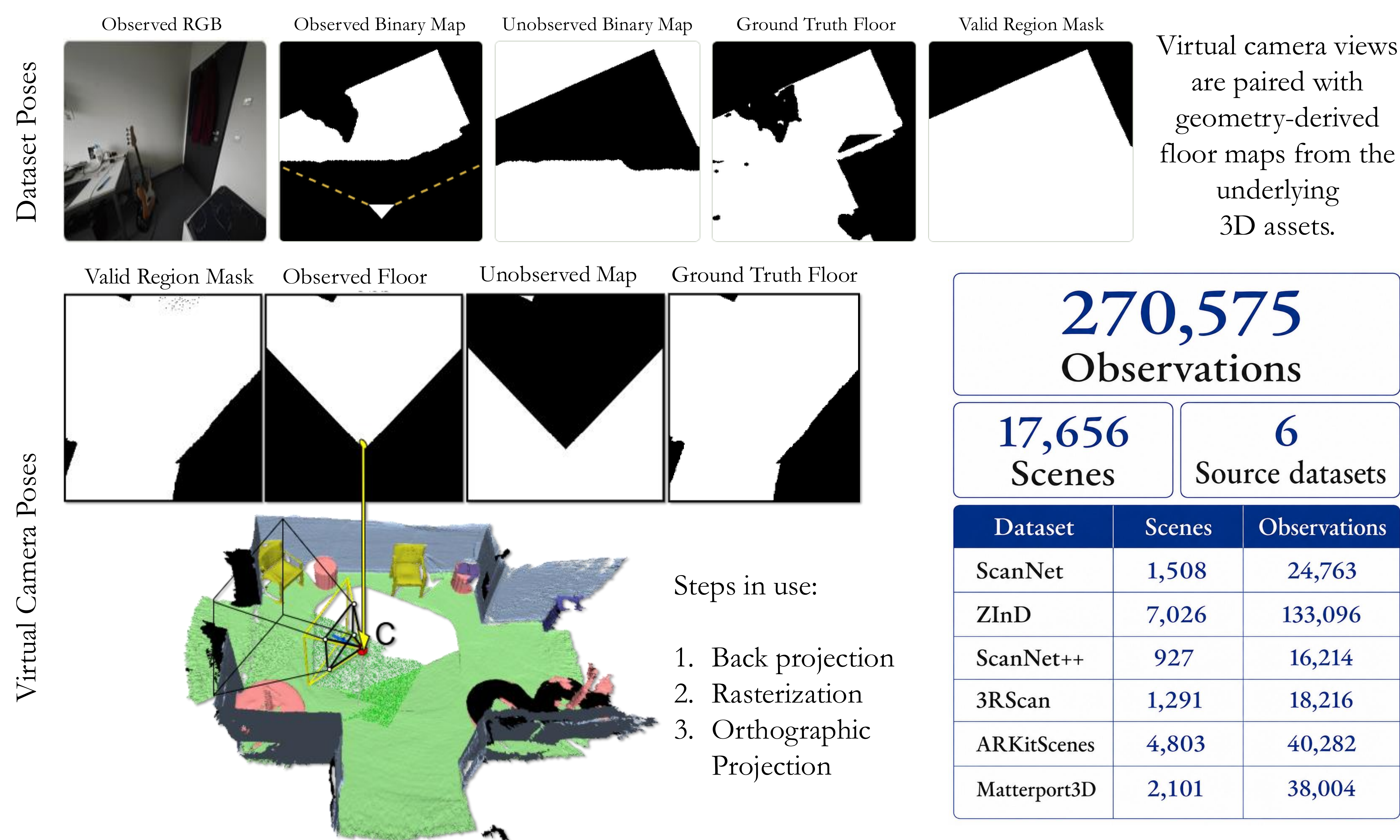


Rather than proposing a new architecture, we systematically benchmark established deterministic and stochastic models for floormap completion.

Benchmark & Results

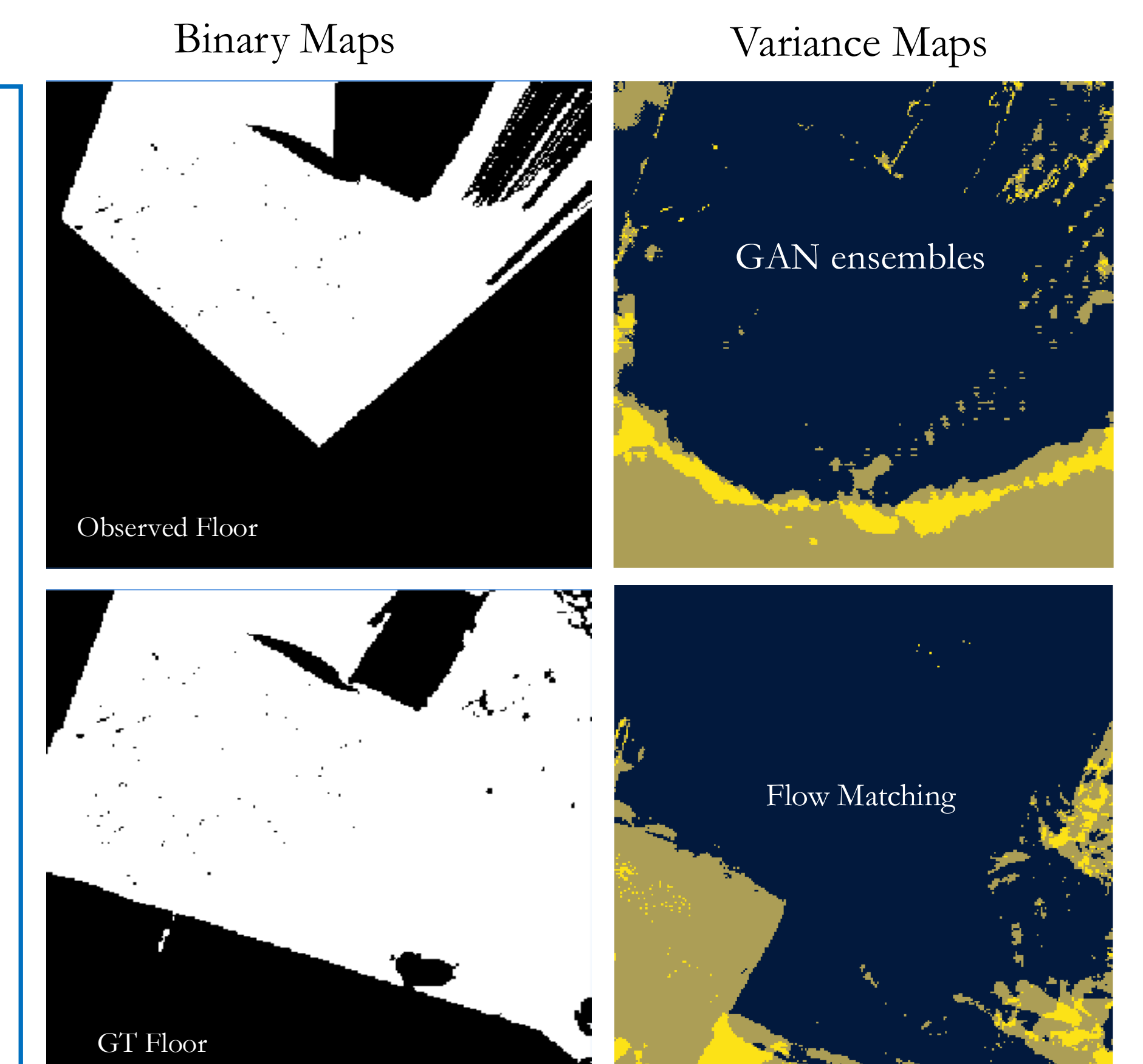


FlatLands Dataset



Main Findings

- Model rankings remain consistent on unseen datasets, showing strong out-of-distribution generalization
- IoU can reward trivial floor guesses; energy scores better capture fidelity and diversity
- Stochastic models preserve plausible alternatives, but severe occlusion erodes their advantage over deterministic models
- Generative models localize uncertainty at ambiguous boundaries; ensembles spread it indiscriminately across the map.
- With RGB input, perception errors dominate and diminish the gains across stochastic completion variants



References

- Datasets:** ZInD'21 · ARKitScenes'21 · Matterport3D'17 · ScanNet'17 · 3RScan'19 · ScanNet++'23
- Baselines:** U-Net'15 · Partial Conv.'18 · LaMa'22
- Generative Methods:** GAN'14 · DDPM'20 · DDIM'21 · Flow Matching'23 · Rectified Flow'23
- Pretrained Models:** Depth Pro'25 · SegFormer'21
- Metrics:** Boundary IoU'21 · Energy Distance'16

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